

2.3 The Precise Definition of a Limit

We now turn our attention to the precise definition of a limit. We replace vague phrases like “gets close to” in the informal definition with specific conditions that can be applied to any particular example.

① An example

Recall

$\lim_{x \rightarrow c} f(x) = L \Leftrightarrow$ as x approaches c from both sides of c ($x \neq c$), $f(x)$ gets closer and closer to L .

Note

- (a) $|a - b|$: Geometrically, it is the distance between a and b .
- (b) $a = b$: $|a - b| = 0$
- (c) $f(x)$ gets closer and closer to L : $|f(x) - L|$ is less than any positive number.
- (d) x approaches c : $|x - c|$ is less than any positive number.
- (*) $f(x)$ gets closer and closer to $L \Leftrightarrow$ In this book, it means that there is no gap between $f(x)$ and L .

To show that the limit of $f(x)$ as $x \rightarrow c$ equals the number L , we need to show that the gap between $f(x)$ and L can be made “as small as we choose” if x is kept “close enough” to c . Let us see what this requires if we specify the size of the gap between $f(x)$ and L .

Example 1 Suppose that $f(x) = 2x - 1$. Then we know that

$$\lim_{x \rightarrow 4} (2x - 1) = 7.$$

However, how close to $c = 4$ does x have to be so that $f(x) = 2x - 1$ is within 2,

0.2 or 0.02 of 7.

Solution We need to find the conditions on x which guarantee that

$$|f(x) - 7| < 2.$$

That is, we need to find a number $\delta > 0$, such that

$$0 < |x - 4| < \delta \Rightarrow |f(x) - 7| < 2.$$

Note that $|f(x) - 7| < 2 \Leftrightarrow |2x - 1 - 7| < 2 \Leftrightarrow |2x - 8| < 2 \Leftrightarrow |x - 4| < 1.$

Thus, when $|x - 4| < 1$, we have

$$|2x - 1 - 7| < 2.$$

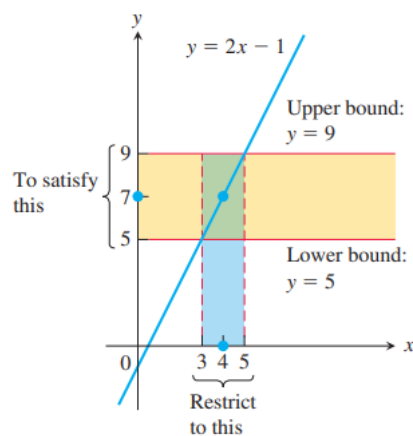
Similarly,

when $|x - 4| < 0.1$, we have

$$|2x - 1 - 7| < 0.2.$$

when $|x - 4| < 0.01$, we have

$$|2x - 1 - 7| < 0.02.$$



If we use ε to replace the small number 2, 0.2, and 0.02, and use δ to replace the number 1, 0.1, and 0.01, we have the following definition.

② Definition of Limit

DEFINITION We write

$$\lim_{x \rightarrow c} f(x) = L,$$

if $\forall \varepsilon > 0, \exists \delta > 0$, such that $0 < |x - c| < \delta \Rightarrow |f(x) - L| < \varepsilon$.

Example 2 Show that $\lim_{x \rightarrow 1} (5x - 3) = 2$.

Solution $\forall \varepsilon > 0$, we need to find a $\delta > 0$, such that

$$0 < |x - 1| < \delta \Rightarrow |5x - 3 - 2| < \varepsilon.$$

Note that $|5x - 3 - 2| < \varepsilon \Leftrightarrow |5x - 5| < \varepsilon$

$$\Leftrightarrow 5|x - 1| < \varepsilon$$

$$\Leftrightarrow |x - 1| < \frac{\varepsilon}{5}.$$

Thus, taking $\delta = \frac{\varepsilon}{5}$, if $0 < |x - 1| < \delta = \frac{\varepsilon}{5}$, then

$$|5x - 3 - 2| = |5x - 5| = 5|x - 1| \leq 5\delta = \varepsilon.$$

So, $\lim_{x \rightarrow 1} (5x - 3) = 2$.

Note How to Find Algebraically a δ for a Given f, L, c , and $\varepsilon > 0$

The process of finding a $\delta > 0$ such that for all x

$$0 < |x - c| < \delta \Rightarrow |f(x) - L| < \varepsilon$$

can be accomplished in two steps.

1. Solve the inequality $|f(x) - L| < \varepsilon$ to get $|x - c| < \rho(\varepsilon)$.
2. Taking $\delta = \rho(\varepsilon)$.

Example 3 For the limit $\lim_{x \rightarrow 5} \sqrt{x - 1} = 2$, find a $\delta > 0$ that works for $\varepsilon = 1$. That

is, find a $\delta > 0$ such that for all x

$$0 < |x - 5| < \delta \Rightarrow |\sqrt{x - 1} - 2| < 1.$$

Example 4 Prove the following results in Section 2.2.

(a) $\lim_{x \rightarrow c} x = c.$

(b) $\lim_{x \rightarrow c} k = k$ (k constant).

Example 5 Prove that $\lim_{x \rightarrow 2} f(x) = 4$ if

$$f(x) = \begin{cases} x^2, & x \neq 2 \\ 1, & x = 2. \end{cases}$$